

SDOC — Shipping Document Check

Averis × Monash Hackathon 2026

Team Claude's Plan · [Member 1] · [Member 2] · [Member 3]

AI-assisted triage of a shipping-documentation inbox, with a deterministic SI-vs-BL check and a human in the loop.

The problem

Averis's documentation team receives hundreds of emails a day. Mixed in are requests to confirm a carrier's **draft Bill of Lading** against the **Shipping Instruction**.

- Today a person opens both files and compares 7 fields by eye
- The same field has different labels on each document: *POD* vs *Discharge Port*, *To the Order of* vs *Consignee*
- One wrong port or container count on a released BL means amendment fees, delays and detention charges

Who it affects: documentation staff, customers waiting on cargo, finance absorbing the fees.

What we built

A pipeline plus a reviewer dashboard (web + Android) that reads a **real mailbox over IMAP** (Gmail, Outlook) or an uploaded `.eml` , as well as the hackathon dataset:

1. **Classify** every email: BL comparison, SI request, invoice query, general, spam
2. **Read** the attachments: txt, PDF, Word, Excel
3. **Extract** the 7 fields, whatever they are labelled
4. **Compare** and name the exact fields that differ
5. **Escalate** anything it cannot decide, with the reason
6. **Draft** the amendment email for a person to send

Results on the 520-email dataset

1.000 organisers' final score (0.30 classify + 0.20 defect-F1 + 0.50 end-to-end)

Metric	Result
Classification accuracy, 5 categories	100 %
Defective BLs caught with the exact wrong fields	46 / 46
False alarms on clean pairs	0
Edge cases escalated with the right reason	20 / 20
Tests	49 passing

Dataset is synthetic, provided by the organisers. Numbers are reproducible with `sdoc run`.

How: AI reads, code decides

```
email -> classify -> gate -> read SI + BL -> extract 7 fields -> compare -> OK / MISMATCH
      rules first,   NEEDS_REVIEW:      heuristics first,   deterministic,
      LLM fallback   missing attachment, LLM fills gaps      unit-tested
                        unreadable, wrong
                        doc type, blank value
```

- Rules and label-synonym matching handle the regular cases for free
- An LLM (Gemini, Claude API or Amazon Bedrock) is a drop-in fallback in two places, never in the comparison
- The comparison cannot hallucinate: ports match on name **and** UN/LOCODE, weights in kg, parties normalised

Architecture

- **Pipeline** (Python): one class per stage behind an abstract base; `build_pipeline()` injects them
- **API** (FastAPI): inbox, results with evidence, reviewer override, live metrics
- **Dashboard** (React + Capacitor): Inbox, Compare, Invoices, Impact, translation; same code ships as a web app and an Android APK
- **Cloud** (AWS Free Plan, \$0): Lambda + Function URL, DynamoDB, S3, Amplify Hosting, Bedrock

Stateless per email → maps straight onto SQS + Lambda workers for scale.

Validation: how we found our own bugs

We measured against the answer key after every change.

Iteration	Score	What was wrong
First run	0.76	"BILL OF LADING INSTRUCTION" typed as a BL; ports compared by code only; bot notices with "Billing" in the subject
Second	0.97	PDF labels physically overlapping values: ConsCigEnReIEeX
Third	1.00	Split PDF lines by font weight

Every fix became a unit test.

Never guesses

Situation	What the tool does
Body says "attached" but nothing is	NEEDS_REVIEW · missing_attachment
Scanned image-only PDF, empty or corrupt file	NEEDS_REVIEW · unreadable
A Commercial Invoice sent as the "BL"	NEEDS_REVIEW · wrong_doc_type
SI field left as ??? or TBA	NEEDS_REVIEW · missing_value
"Please send the draft BL" (nothing to compare yet)	OK, no comparison

A blank is uncertainty, not a discrepancy. The reviewer sees the reason and the evidence line for every value.

Live demo

1. Inbox: 520 emails triaged, filter to MISMATCH
2. Open one: SI and BL side by side, two fields highlighted, hover shows the source line
3. Copy the drafted amendment email
4. Open a NEEDS_REVIEW case: wrong document type, with the reason
5. Override a result as a reviewer; audit trail records who
6. Impact panel: live metrics from `/metrics`

Impact and roadmap

Now: minutes per email → seconds; zero false alarms means reviewers trust the queue.

Next (finals): Gmail add-on / Outlook add-in · SQS workers · OCR for scanned PDFs · learn new label synonyms from reviewer corrections · Outlook add-in.

Measures of success: end-to-end catch rate, false-alarm rate, % auto-handled vs escalated, reviewer minutes saved.

Challenges

- Same information, different labels and file formats
- PDF text layers that interleave label and value glyphs
- Deciding *when not to decide*: distinguishing "send me the draft" from "the attachment was dropped"
- Keeping the cloud bill at \$0

Thank you

Repo: github.com/VRP2206/Averis_hackathon_2026

Demo: [URL]

Student prototype for the Averis × Monash Hackathon 2026. Not an Averis product.